

Information Technology & AI

Project Name: Employee Check-In System





The Problem: In offices and companies, keeping track of employee attendance can be difficult when it is done manually. Employees may forget to sign attendance registers, and records can sometimes contain mistakes. We can solve this problem by building a smart check-in system that automatically records employee login and logout using RFID cards.

🎯 **What You Will Design:** Students will build a smart employee check-in system that uses RFID cards for attendance. Employees can scan their RFID cards to log in when entering the office and scan again to log out when leaving.

📡 What are RFID Cards and Tags?

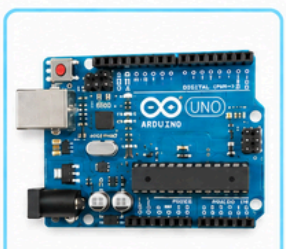


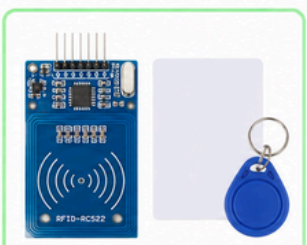
RFID cards and tags contain a small electronic chip that stores unique information. An RFID reader can read this information wirelessly when the card is placed nearby.

Example: 🏢 Many offices and schools use RFID ID cards to record attendance and control entry into buildings.



Components Required

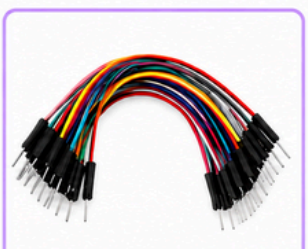

Arduino Uno


RFID Card and Tags



OLED Display

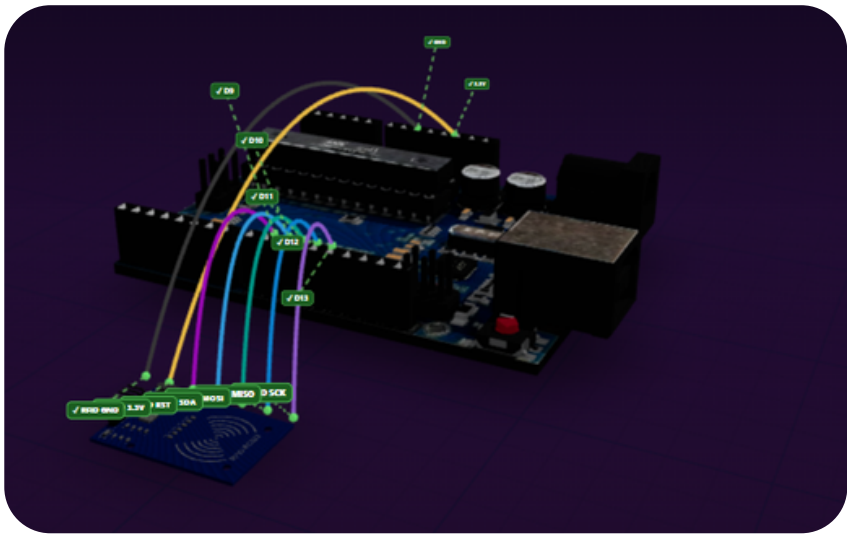

LED


Jumper Wires

Circuit Designing

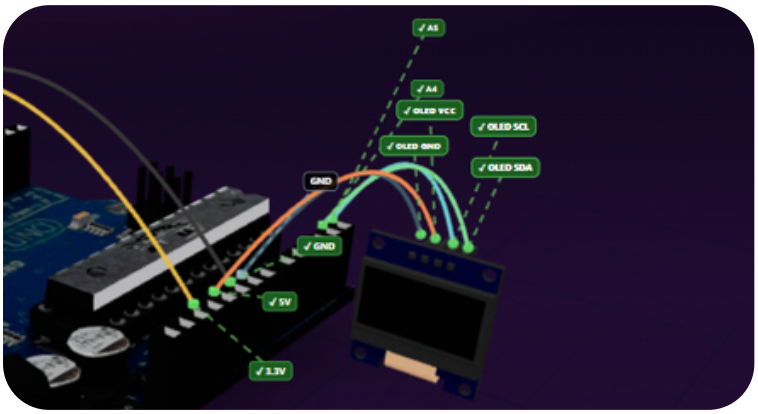
Step 1: Connect the 4x4 keypad module to the Arduino Uno.

Sensor Pins	Arduino Pins
SDA Pin	Digital Pin 10
SCK Pin	Digital Pin 13
MOSI Pin	Digital Pin 11
MISO Pin	Digital Pin 12
RST Pin	Digital Pin 9
3.3V Pin	3.3V
GND Pin	GND



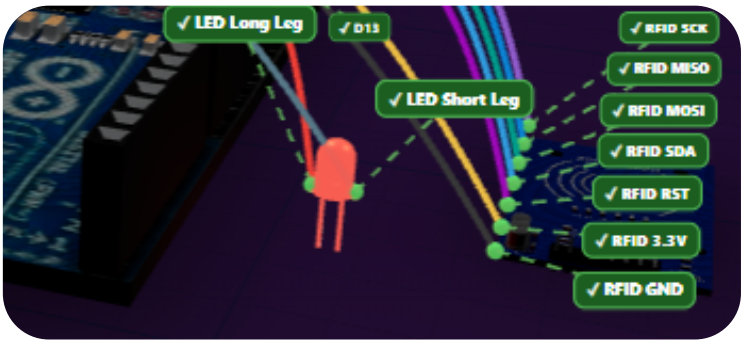
2. Connect the OLED display to the Arduino Uno.

OLED Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
SDA Pin	Analog Pin A4
SCL Pin	Analog Pin A5



3. Connect the LED to the Arduino Uno.

LED Pin	Arduino Pins
Long Leg	Digital Pin 7
Short Leg	GND Pin



Project Coding

1. Open the Arduino Uno software to add the OLED and RFID reader libraries.

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
```

2. Make code to define the pins RFID and OLED pins.

```
#define SS_PIN 10
#define RST_PIN 9
#define LED_PIN 7

MFRC522 rfid(SS_PIN, RST_PIN);

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

bool loggedIn = false;
```

3. Make code initialize the OLED and RFID reader.

```
void setup()
{
    Serial.begin(9600);

    SPI.begin();
    rfid.PCD_Init();

    pinMode(LED_PIN, OUTPUT);

    display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
```

4. Make code to set the text size for the OLED display.

```
    display.clearDisplay();
    display.setTextSize(1);
    display.setTextColor(WHITE);

    display.setCursor(10, 25);
    display.println("Scan RFID Card");
    display.display();
}
```

5. Make code to read the RFID Card

```
void loop()
{
    if (!rfid.PICC_IsNewCardPresent())
        return;

    if (!rfid.PICC_ReadCardSerial())
        return;

    digitalWrite(LED_PIN, HIGH);

    display.clearDisplay();
```

6. Make code to use the conditions to check for the new RFID cards.

```
if (loggedIn)
{
    display.setCursor(10, 20);
    display.println("WELCOME");
    display.setCursor(10, 40);
    display.println("Login Successful");
    loggedIn = true;
}
else
{
    display.setCursor(10, 20);
    display.println("GOODBYE");
    display.setCursor(10, 40);
    display.println("Logout Successful");
    loggedIn = false;
}
```

7. Make code to display the Login Status Message

```
display.display();

delay(2000);

digitalWrite(LED_PIN, LOW);
```

8. Make code to display the "Scan RFID Card" Message

```
    display.clearDisplay();
    display.setCursor(10, 25);
    display.println("Scan RFID Card");
    display.display();

    rfid.PICC_HaltA();
}
```

Project Output





🌾 Where Is This Used in Real Life?

RFID attendance systems are used in offices, schools, factories, and company buildings. Employees scan their cards to record attendance and enter secure areas.

Learning Outcomes

- The students will be able to use RFID cards and readers for identification and attendance.
- The students will be able to display login and logout messages on an OLED screen.
- The students will be able to build an automatic employee check-in and check-out system.



Section 1: Multiple Choice Questions

1. What problem does Riya want to solve?

a) Employees need new computers
b) Manual attendance can have mistakes
c) The office needs more chairs
d) Employees need new ID cards

2. Which component is used to identify an employee?

a) LED
b) OLED Display
c) RFID Reader and Tag
d) Jumper Wire

3. What happens when an employee scans a valid RFID card?

a) Attendance is recorded
b) The Arduino turns off
c) The RFID card is erased
d) The OLED stops working

Section 2: True and False Statements

1. Employees may forget to sign a manual attendance register.....

2. An RFID card can be read wirelessly by an RFID reader when it is placed nearby.....

3. The OLED display is used to identify the employee's RFID card.....